

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] DEC 2025 - JAN 2026

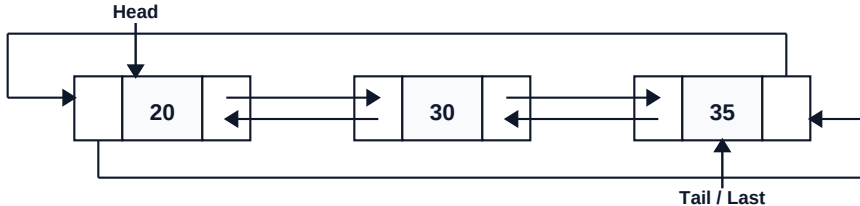
Paper Code: AIDS/AIML/IOT-201

Subject: Data Structures

Time: 3 Hours

Maximum Marks: 60

Note: Attempt all questions as directed. Internal choice is indicated.

Q. No.	Question	Marks
Q1 (a)	Write an algorithm for Depth-First Search (DFS). Give an example.	5
(b)	Convert the infix expression $(A + B) * (C - D^E) / H$ to postfix notation. Show all the steps.	5
(c)	An array $X[20]$ starts at base memory address 1000. Each element takes 4 bytes. Find the address of $X[10]$ using row-major representation.	5
(d)	Write an algorithm to insert a node at the beginning of a linked list.	5
(e)	A hash table of size 10 stores 6 elements. If linear probing is used, what is the load factor (α)? What happens if $\alpha > 1$?	5
(f)	Differentiate between Eulerian and Hamiltonian paths with examples.	5
(g)	Construct a Binary Tree using the following In-order and Pre-order traversals of a tree: INORDER: D G B A H E I C F PREORDER: A B D G C E H I F	5
(h)	What is a Complete binary tree? If the number of nodes in a complete tree is 100000, find its depth.	5
Q2 (a)	A circular queue of size 5 initially empty is given. If the following operations are performed: Insert(11), Insert(23), Insert(34), Delete(), Insert(45), Insert(50), Insert(60) . Show the front and rear positions after each operation.	7
(b)	What is Algorithmic Complexity, and explain its importance?	3
OR		
Q3 (a)	Implement a Queue using arrays and demonstrate enqueue and dequeue operations with the help of an example. What are its various applications?	6
(b)	Give the features of various data structures such as List, Set, Tuple and Dictionary. Give an example of each.	4
Q4 (a)	Sort the array $A = [29, 10, 14, 37, 13]$ using Quick Sort with last element as pivot. Write the sequence of swaps and the final sorted output.	5
(b)	What do you mean by time and space trade-off? Define various asymptotic notations and derive the Big O notation for linear search.	5
OR		
Q5 (a)	Perform Merge Sort on array $A = [13, 12, 14, 6, 7, 8]$. Show how the array is divided and merged in a step-by-step manner.	5
(b)	Consider a circular doubly linked list as below:  Perform the following operations and show the procedure: 1) Insert the node 10 at the beginning 2) Delete node at the end	5

[P.T.O.]

Q. No.	Question	Marks
Q6 (a)	Construct a Binary Search Tree (BST) by inserting the following sequence of keys in order: 45, 12, 77, 34, 55, 94, 25, 50. Then, delete 55 and show the updated tree.	7
(b)	Explain how tree balancing affects search performance.	3
OR		
Q7 (a)	Explain how in-order traversal can be done without recursion or stack using a threaded binary tree. Draw an example with 5 nodes.	5
(b)	An AVL Tree has the sequence [60, 70, 80, 90, 100] . Show rotations after each insertion and compute height after final insertion.	5
Q8 (a)	For graph: V = {A, B, C, D, E} , Edges = {A→B=2, A→C=3, B→D=5, C→D=1, D→E=2} . Find the shortest path from A to E using Dijkstra's Algorithm.	6
(b)	Explain how indexing improves file access time. What is the difference between dense and sparse indices?	4
OR		
Q9 (a)	A file contains 1000 records, each record takes 100 bytes, and the block size = 1 KB. Compute: 1. Blocking factor (bfr) 2. Number of blocks required	4
(b)	Using hash function $h(k) = k \bmod 7$, insert keys {10, 20, 15, 33, 47, 28} . Resolve collisions using Linear probing and show the final hash table.	6
